

LESSON 11.5

SCATTER PLOTS AND LINES OF FIT



LESSON INTRODUCTION

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

MA.8.DP.1.3 Given a scatter plot with a linear association, informally fit a straight line

LEARNING TARGETS

- I can describe patterns of associations in data on a scatter plot
- I can informally fit a straight line to data on a scatter plot.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

WORKING TOWARD

MA.912.DP.1.2 Interpret data distributions represented in various ways, state whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.

MA.912.DP.1.3 Explain the difference between correlations and causation in the contexts of both numerical and categorical data.

ESSENTIAL QUESTIONS

- How do you determine an appropriate line of fit for a scatter plot?
- What does the information in a scatter plot tell me?
- What is the purpose of a line of fit?
- How is it possible to have multiple different answers that are correct?

LESSON NARRATIVE

Students continue their interpretation of real-world scatter plots with a collaboration and an activity. Students informally fit a straight line to linear data. Students then are given the opportunity to work in pairs to interpret the plots given to them, draw lines of fit, and then compare their work to another group that had the exact same scatter plot.

COMPONENT SUMMARY

11.5.1 WARM-UP (~5 MIN.)

Bell work to refresh the idea of slope and intercept interpretation

11.5.3 ACTIVITY (10-15 MIN.)

Students use a straightedge to practice drawing lines of fit given different sets of data

11.5.2 COLLABORATION (15-20 MIN.)

Collaborate to identify patterns or relationships produced by data in different scatter plots

11.5.4 WRAP-UP (~5 MIN.)

Formative assessment on drawing a line of fit

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- 11.5.3 scatter plots
- (Optional) Plastic sleeves
- Straightedge (ruler, spaghetti, etc.)
- (Optional) Graphing utilities for lines of fit

ADDITIONAL PREPARATION

- Consider displaying visuals of key terms from Unit 8 that show the different relationships, associations, etc.
- Consider grouping students based on data from the previous lessons on features and associations of scatter plots for 11.5.2 Collaboration.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may misunderstand that end behavior is looked at from both sides of the line.
- Students may misinterpret negative versus positive associations.
- Students may misinterpret a line of fit as having to go through all points on the scatter plot.

THINGS TO CONSIDER

- This lesson extends vocabulary and concepts covered in previous lessons. Continuous use of the vocabulary is helpful to student understanding.
- Students are working on describing associations and connecting them to the slope of their lines of fit.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong?

In 11.5.2 Collaboration, question 1b is specifically designed to have students choose which scatter plot does not belong. Before moving on to 11.5.3 Activity, consider discussing student choices and their justifications to allow deeper understanding of associations and relationships concerning scatter plots.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K.12.MTR.5.1 Patterns and Structure

Within 11.5.2 Collaboration, students take previously learned terms and are asked to use them to describe the patterns of data within the scatter plots. Furthermore, in 11.5.3 Activity, students explore how these patterns of data can be represented in a line of fit. Students are also asked to compare and contrast their work with another group's work as a way of extending the concepts that were featured in this lesson.

DIFFERENTIATE INSTRUCTION

Collaboration and discussion are main components of learning in this lesson. Utilize sentence frames to enhance student discussion and target key learning opportunities as students collaborate throughout the lesson. Sentence frames provide low-floor, high-ceiling entry points for learners to either enter the conversation or extend their thinking. This can be particularly impactful for ELL students. For example, consider using the following frames during 11.5.2 Collaboration:

- Scatter Plot ____ does not belong because _____.
- I would describe the linear association of Scatter Plot ____ as weak/strong because _____.
- My line is similar to my partner's because _____. Our lines are different because _____.

11.5.1 WARM-UP: BELL WORK

ENRICHMENT

To help students recall determining slope:

- What are two points that are on the line that we can use to determine their coordinates?
- How can we use these two points to determine the slope?

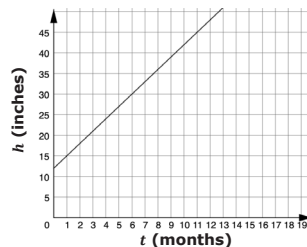
QUESTIONING

For students who finish early, consider asking them to take the information from the graph to write the equation of the line in slope-intercept and standard form.



11.5.1 Warm-Up: Bell Work

1. The graph shows the height in inches, h , of a bamboo plant t months after it has been planted.

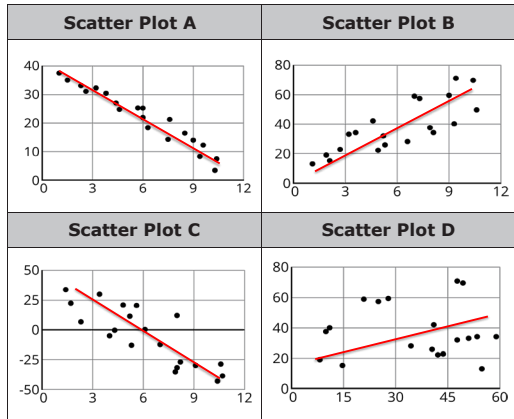


- What is the slope of the line?
 $\frac{30}{10}$ or 3
- Interpret the slope in this context.
Each month the bamboo plant grows 3 inches.
- The y -intercept of the line is 12. Interpret the meaning of the y -intercept in context.
The bamboo plant was 12 inches tall when planted.



11.5.2 Collaboration: Patterns of Association

1. Four scatter plots are shown.



a. Describe the direction of the linear association between the two variables in each scatter plot.

Scatter Plot	Direction of the Association
A	Negative
B	Positive
C	Negative
D	Positive

- b. Explain which of the scatter plots does not belong based on the strength of the linear association between the two variables.
Answers may vary.
Sample response: Scatter Plot D does not belong because it has a weak linear association.
- c. Discuss with your partner whether any of the scatter plots appear to have any outliers.
Answers may vary.
Sample response: None of the scatter plots have any outliers.
- d. Use a straightedge to draw a line that models the linear relationship between the two variables on each scatter plot.
Answers will vary.
- e. Compare your lines with your partner's. Discuss any differences and make revisions to your lines, if necessary.
Answers will vary.
- f. Explain which of the scatter plots does not belong based on the y-intercept of the lines you drew to model the linear relationships.
Answers will vary.
Sample response: Scatter Plot D does not belong because the y-intercept falls in the middle of the y-axis whereas the rest of the scatter plots the y-intercept falls either at the top or bottom of the y-axis.
- g. Explain which of the scatter plots does not belong based on the slope of the lines you drew to model the linear relationships.
Answers will vary.
Sample response: Scatter Plot D has the flattest slope.

11.5.2 COLLABORATION: PATTERNS OF ASSOCIATION

ENRICHMENT

For Scatter Plot D, because the association is weak, some students may not recognize there is a positive association between the variables in Scatter Plot D. Consider listening/looking for this as students work through this collaboration with their partner. Guidance can be offered as:

- How does the plot of data in Scatter Plot D compare to Scatter Plot A?
- What do you notice about the distance between points?

TEACHER MOVES

Which One Doesn't Belong?

For 1b, consider opening this for discussion before completing the Activity in the next section in order to deepen student understanding and discussion about associations and patterns in scatter plots.

ENRICHMENT

For 1c, teachers should listen to student discussion. None of them appear to have any outliers, but students may think that Scatter Plot D does. Depending on how students engage with the Scatter Plots A-C, consider engaging in a discussion regarding Scatter Plot D at the conclusion of the collaboration.

TEACHER MOVES

The purpose of question 1f is to make connections to linear relationships/functions, and to recognize the features they explore, and to apply to the lines used to model relationships in scatter plots as well. They'll delve deeper into this concept in future lessons in the unit as well as in Algebra 1.

11.5.3 ACTIVITY: INFORMALLY FITTING LINES TO DATA

DIFFERENTIATE INSTRUCTION

To provide additional entry points, consider:

- If possible, provide clear rulers or straightedges to help students make their lines of fit confidently
- If possible, put the scatter plots in sheet protectors so students can use white board markers to draw their lines of fit.
- If that's not possible, consider printing multiple scatter plots to a page so that each copy is only $\frac{1}{4}$ of a page.
- If digital devices are available, consider having students draw a line of fit with the scatter plots using a spreadsheet application.

TEACHER MOVES

Before concluding the activity, consider discussing what makes a line of fit better fit the data given. For example, was it by how many points the line went through, or how close it was to the beginning or ending data points?



11.5.3 Activity: Informally Fitting Lines to Data

1. Your teacher will give you and your partner a scatter plot and information about the relationship it models.

- a. Record the letter of the provided scatter plot.

Answers will vary.

Scatter Plot _____

- b. Describe any patterns of association.

Answers will vary.

Sample Response: the scatter plot that we looked at has a positive association.

- c. Work with your partner to use a straightedge to draw a line that models the relationship between the two variables on the scatter plot.

Answers will vary.

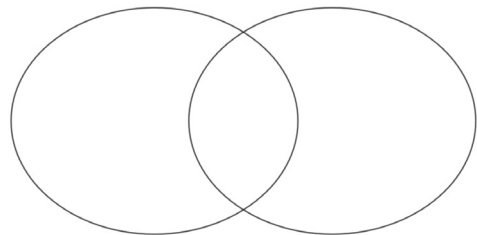
- d. Explain how the slope of your line of fit relates to the patterns of association described in part b.

Answers will vary.

2. Find a group with the same lettered scatter plot as you.

- a. Compare and contrast both groups' work. Record your observations on the Venn diagram.

Answers will vary.



- b. Both groups had the same scatter plot. How would you explain any differences in your work?

Answers will vary.

Sample response: Even though we had the same scatter plot, our lines of fit were not exactly the same and had different slopes.

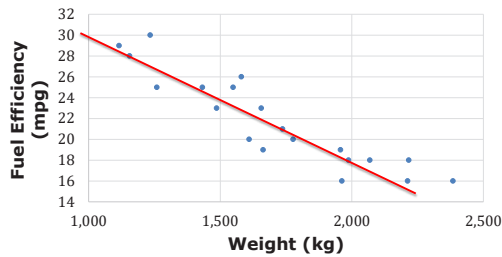
- c. Explain which group's line fits the data better.

Answers will vary.



11.5.4 Wrap-Up: Ticket Out the Door

1. The scatter plot shows the relationship between a car's weight in kilograms (kg) and its fuel efficiency in miles per gallon (mpg).



- a. Informally fit a line to the data using a straightedge.
Answers will vary.
- b. Describe how the slope of your line relates to the strength and direction of the association between the weight and fuel efficiency of a car.
Answers will vary.
Sample response: Based on the slope of the line the relationship between fuel efficiency and the weight of a car have a relatively strong negative association.

11.5.4 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Consider providing a word bank or sentence starters for 1b to assist students in articulating their interpretation of the line of fit.

QUESTIONING

For students who may finish early, consider asking them to interpret what that negative association means. Sentence starter such as, "The heavier a car is..." might prove helpful. This offers students a preview of coming lessons.